COURSE LOGISTICS AND EXPECTATIONS

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CEVE 543

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BY THE END OF THIS COURSE, YOU WILL BE ABLE TO...

1. Frame an engineering or financial decision as a risk analysis problem, and name the quantities that have to be estimated.
2. Fit extreme value models, analyze time series, pool information across sites, and generate event sets, carrying the uncertainty through to the answer.
3. Choose a method, defend the choice, and explain its assumptions to someone who is not a specialist.
4. Use, and choose not to use, AI tools in your learning and in your analysis.

COURSE ORGANIZATION

Module	Weeks	Driving question
Foundations	1-3	Why do we care about quantitative representations of climate hazard?
Univariate extremes	4-7	How do we characterize the distribution of one hazard at one location, and with what uncertainty?
Spatiotemporal extremes	8-14	How do we sample hazards across space and time to characterize complex extremes?
Synthesis	15	

GRADE

Component	Weight
Participation	10%
Labs	10%
Written tests (2)	30%
Oral examinations (2)	20%
Final project	30%

There is no final exam; we will use the final examination slot for final presentations.

WRITTEN TESTS

Questions will parallel weekly practice problems (ungraded) and labs.

Closed book and no calculator; some equations may be provided.

Test	Date	Covers
Test 1	Fri Oct 16	Weeks 1-7
Test 2	Mon Nov 30	Weeks 8-14

ORAL EXAMINATIONS

One after each written test, about 15 minutes each. **We will practice these.**
Orals are recorded so grading can be audited for consistency.

Orals	When
Format, rubric, and a demonstration oral	Fri Sep 18, in class
Oral 1, weeks 1-7	By signup, weeks 9-10
Oral 2, weeks 8-14	By signup, week 15

LABS

1. Do setup *before* in-class lab
2. We will work through labs together
3. Labs are due on Canvas as a rendered PDF plus a link to your repository

Points	What it means
3	A thoughtful answer
2	A quick answer
1	You turned it in
0	You did not turn it in

I grade the answers you write, not whether your code matches mine. Two lowest dropped. Labs assume you use an LLM, and they use Claude Code.

FINAL PROJECT

Either:

1. Research a method the course does not cover and compare it against alternatives.
2. Learn a software package that answers a question like the ones we cover, then use methods from the course to assess it.

PARTICIPATION

Judged on preparation and effort, not on correctness.

- Attend, and tell me when you cannot.
- Do the reading and submit your reading questions.
- Answer as best you can when called on.

DEVICES

- Please don't use laptops for note taking (fine for labs!)
- Please stay focused; avoid devices except for polling
- Recording class is not permitted, including transcription tools

THERE IS NO BLANKET BAN ON AI

1. Shared responsibility. I commit to teaching you; you commit to learning. Using AI to bypass thinking costs you the skills you are paying for.
2. Assessment design. Most of the grade comes from oral examinations and in-person written tests, so there is no incentive to turn in work a model wrote.
3. Open dialogue. We talk in class about how we are using these tools, which means you can say what you did without being shamed for it.

WHAT IS THE BEST THING YOU HAVE USED AI FOR?

Set your Poll Everywhere screen name to your Rice NetID so your answers are recorded.

PollEV.com/jdossgollin

WHAT IS THE WORST THING YOU HAVE USED AI FOR?

Set your Poll Everywhere screen name to your Rice NetID so your answers are recorded.

PollEV.com/jdossgollin

EVERYTHING ELSE IS ON THE COURSE WEBSITE

https://ceve543.github.io/